

George Challenges the Dragon (gcd)

Evil George has kidnapped the princess. Now Drogo the dragon comes to rescue her. George prefers challenging the dragon intellectually rather than fighting him, so he tells Drogo:

“I will free the princess if you can solve the following puzzle. I tell you two positive numbers N and D , and you have to tell me two different positive N -digit numbers A and B so that their greatest common divisor is D .”

Can you help the dragon to solve this challenge and save the princess? Beware, George might be so evil that there is no possible answer!

Note: the greatest common divisor of two positive integer numbers A and B is the greatest integer k such that both A and B are multiples of k .



Figure 1: The dragon comes to rescue the princess.

 Among the attachments of this task you may find a template file `gcd.*` with a sample incomplete implementation.

Input

The input has two lines, both of them contain a single integer. The first line contains N , the second line contains D .

Output

You need to write two different positive integers A and B separated by a space. Both A and B must have N digits and their greatest common divisor must be D . If there are no such numbers, you should write `0 0` to the output. If there are multiple possible solutions, you can print any of them.

 A and B may not fit into a 32 bit integer. To avoid the *integer overflow* error, use C/C++ `long long` or Pascal `Int64`.

Constraints

- $1 \leq N \leq 18$.
- $1 \leq D \leq 10^9$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (12 points) $N \leq 3$.

- **Subtask 3** (24 points) $N \leq 7$.

- **Subtask 4** (10 points) $D = 1$.

- **Subtask 5** (15 points) $D \leq 100$.

- **Subtask 6** (39 points) No additional limitations.


Examples

input	output
3 9	180 729
6 666666	0 0

Explanation

In the **first sample case**, Drogo has to tell two different 3-digit numbers whose greatest common divisor is 9. There are many possible solutions, e.g. 180 and 729.

In the **second sample case**, there is only one 6-digit number which is divisible by 666666, it is 666666 itself, which means that it is not possible to give two different 6-digit numbers whose greatest common divisor is 666666.